sheet_Location.ino

```
#include <SoftwareSerial.h>

#include <TinyGPS.h>
#include <ESP8266WiFi.h>
#include <WiFiClientSecure.h>

#define ON_Board_LED 2  //--> Defining an On Board LED, used for indicators when the process of connecting to a wifi router

/* This sample code demonstrates the normal use of a TinyGPS object.
   It requires the use of SoftwareSerial, and assumes that you have a
   4800-baud serial GPS device hooked up on pins 4(rx) and 3(tx).
*/

TinyGPS gps;
SoftwareSerial ss(4, 2);

const char* ssid = "JioFiber-4G"; //--> Your wifi name or SSID.
const char* password = "1234567890"; //--> Your wifi password.


//----------------------------------------Host & httpsPort
const char* host = "script.google.com";
const int httpsPort = 443;
//----------------------------------------

WiFiClientSecure client; //--> Create a WiFiClientSecure object.
//https://script.google.com/macros/s/AKfycbxOsEA67mi8tfY9PfvSbbhz9qaHvoIhzlseHizzbiP7EaV1z7cc-gDdg59_42BeqNXzvw/exec?humidity=2
String GAS_ID = "AKfycbzawvSYbPRQgqNAP4fTMCIDY4yFh4E1oPMNkyYdVH18SXZ5Ii60-YiC5IWwIO_Zkxps"; //--> spreadsheet script ID
//https://script.google.com/macros/s/AKfycbzawvSYbPRQgqNAP4fTMCIDY4yFh4E1oPMNkyYdVH18SXZ5Ii60-YiC5IWwIO_Zkxps/exec
//https://script.google.com/macros/s/AKfycbzawvSYbPRQgqNAP4fTMCIDY4yFh4E1oPMNkyYdVH18SXZ5Ii60-YiC5IWwIO_Zkxps/exec?destination=28.50124,77.09610
//https://script.google.com/macros/s/AKfycbzawvSYbPRQgqNAP4fTMCIDY4yFh4E1oPMNkyYdVH18SXZ5Ii60-YiC5IWwIO_Zkxps/exec?location=28.50124,77.09610
//AKfycbzawvSYbPRQgqNAP4fTMCIDY4yFh4E1oPMNkyYdVH18SXZ5Ii60-YiC5IWwIO_Zkxps
    float flat, flon;

void setup()
{
  Serial.begin(115200);
  ss.begin(4800);

  Serial.print("Simple TinyGPS library v. "); Serial.println(TinyGPS::library_version());
  Serial.println("by Mikal Hart");
  Serial.println();

```

Fig. 4: Code snippet

```
        newData = true;
      }
    }

  if (newData)
  {
    //float flat, flon;
    unsigned long age;
    gps.f_get_position(&flat, &flon, &age);
    Serial.print("LAT=");
    Serial.print(flat == TinyGPS::GPS_INVALID_F_ANGLE ? 0.0 : flat, 6);
    Serial.print(" LON=");
    Serial.print(flon == TinyGPS::GPS_INVALID_F_ANGLE ? 0.0 : flon, 6);
    Serial.print(" SAT=");
    Serial.print(gps.satellites() == TinyGPS::GPS_INVALID_SATELLITES ? 0 : gps.satellites());
    Serial.print(" PREC=");
    Serial.print(gps.hdop() == TinyGPS::GPS_INVALID_HDOP ? 0 : gps.hdop());

    Serial.println("==========");
  Serial.print("connecting to ");
  Serial.println(host);

  //----------------------------------------Connect to Google host
  if (!client.connect(host, httpsPort)) {
    Serial.println("connection failed");
    return;
  }
  //----------------------------------------
```

Fig. 5: Capturing and sending the GPS data to Google Sheet

the circuit accordingly. Fig. 1 depicts the GPS device attached to a shipment, and Table 1 lists the bill of materials used in this device.

Circuit diagram

Fig. 2 shows the simple circuit diagram for GPS path tracking for logistics. It is basically the interconnection of AIR 530 GPS and ESP chip 8285/8266.

First, create a Google Sheet for the shipment, cargo list, and tracking of all the cargo. Then follow the instructions for setting up the Google Sheet app. You can refer to the author's previous article titled "IOT Data Logging Using Google Sheets," which can also be seen at

https://www.electronicsforu.com/electronics-projects/iot-data-logging